TAFS API Documentation

Bill Inquiry & Bill Payment Integration
for United Bank Limited (UBL)

The American Foundation School (TAFS)
TAFS IT and Software Department

Aawaiz Ali

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Open Questions and TAFS Decisions at a Glance

Open questions for UBL

Please answer these before integration testing. TAFS will adapt to UBL's answers.

1. **Are 15-digit 1Bill consumer numbers acceptable?** (10-digit prefix + 5-digit CC.) If 1Bill requires a different length, please state it.
2. **Shall TAFS give 1Bill a separate endpoint?** TAFS can provide dedicated 1Bill endpoints (e.g. `/api/v1/ubl/onebill/bill-inquiry` and `/bill-payment`) that always add the 1Bill charges, instead of relying on `channel=ONEBILL` in the request. Which does UBL prefer? If the same endpoint is used, can UBL always send `channel=ONEBILL` on 1Bill inquiries?
3. **What are the 1Bill charges?** Flat amount or percentage, any slabs or caps, and whether they differ by paying bank or wallet. This document assumes PKR 30 until UBL confirms.
4. **1Bill settlement.** Please confirm that 1Bill collects its charges from the payer and keeps them, and that only the fee amount is settled into the TAFS account.
5. **Please issue TAFS's 10-digit 1Bill prefix** (Section 2.2) and confirm when it will be live.
6. **Payment method tracking.** Can UBL send `payment_method` on every payment, for every channel (Section 7.2)? For 1Bill, can UBL also send the paying bank or wallet?
7. **OTC search by CC.** Can the branch teller screen search by CC and show the inquiry result before accepting payment?
8. **Cheques.** Are cheques posted to TAFS when deposited or only once cleared? If at deposit, how will TAFS be told about a returned cheque?
9. **Reversals.** This version has no reversal API; errors are corrected by the school on request. Does UBL require an online reversal message? If so, please share its format.
10. **Daily reconciliation.** Can UBL share a daily settlement statement (transaction id, amount, channel, payment method)?
11. **Message format.** Does UBL require its own standard message format (field names, fixed-length amounts, response codes)? If so, please share the specification.
12. **Testing and security.** Please share UBL's outbound IP addresses (for allowlisting), a technical contact, and the preferred UAT window. TAFS will provide test students and a UAT base URL.

TAFS has decided

1. **CC numbers are 5 digits, zero-padded.** No CC is shorter than 4 digits today, and CC 10,000 is close, so TAFS writes every CC as 5 digits with a leading zero (e.g. `08088`). This is how it is printed on UBL vouchers. Leading zeros are optional when entered (`08088` = `8088`).
2. **1Bill consumer number = 15 digits:** the 10-digit 1Bill prefix + the 5-digit CC (e.g. `1000000000` + `08088` = `100000000008088`).
3. **Search by CC returns the student's most recent voucher.** If a student has **two active vouchers**, the **most recent one is shown and collected first**. The older one is presented only after the most recent one has been paid. TAFS consents to this method (Section 3).
4. **The parent bears the 1Bill charges.** They are added on top of the fee shown to the payer; TAFS books only the fee amount.
5. **Exact amount only, with mandatory channel and payment method** on every payment.

Purpose of This Document

This document describes the interface TAFS provides to United Bank Limited (UBL) for collecting school fees through UBL channels. It covers:

- the collection channels and how a payer identifies a student on each;
- the **Bill Inquiry** (`GET`) and **Bill Payment** (`POST`) APIs, with request and response samples for success and failure;
- the amount rules: within due date, after due date, and the 1Bill charges, which the payer bears;
- mandatory tracking of the **payment channel** and **payment method** on every payment;
- TAFS's stated consent to resolving every search by CC to the student's most recent voucher;
- the open questions TAFS asks UBL to answer, listed first, right after the cover page.

Every payload in this document matches what the TAFS implementation sends and accepts.

1 Collection Channels

Channel	How the payer pays	Identifier entered
UBL branch (OTC)	Cash, cheque or pay order at any UBL branch counter.	The student's **CC number**.
UBL digital app	Bill Payment → select the **TAFS** beneficiary.	The student's **CC number**.
UBL internet banking	Bill Payment → select the **TAFS** beneficiary.	The student's **CC number**.
1Bill (any bank / wallet)	"1Bill Invoices" option in any participating bank or wallet.	The **1Bill consumer number** (Section 2.2).

These instructions are printed at the bottom of every TAFS fee voucher issued for collection into the UBL account.

2 Identifying a Student

2.1 CC number

Every TAFS student has a **CC number**, a numeric student code. To stay unambiguous as CCs reach 10,000, TAFS treats every CC as **5 digits, zero-padded**, and prints it that way on UBL vouchers (e.g. `08088` for CC 8088). The CC is the identifier for OTC, the UBL app and internet banking. Leading zeros may be sent or omitted (`08088` and `8088` identify the same student).

2.2 1Bill consumer number

For 1Bill, the consumer number is a fixed **15-digit** number:

```
[10-digit TAFS 1Bill prefix][CC as 5 digits, zero-padded]
```

Examples, with prefix `1000000000`:

- CC `8088` → `100000000008088`
- CC `10001` → `100000000010001`

Request to UBL: 10-digit 1Bill prefix. TAFS requests UBL to issue (or arrange through 1Bill) the **10-digit 1Bill prefix** for The American Foundation School and share it with TAFS. Until it is received, TAFS vouchers and this document use the placeholder `1000000000`. Once UBL shares the prefix, TAFS will switch to it by configuration; no code change is needed on either side.

2.3 Voucher number

Vouchers collected through UBL carry a plain serial voucher number (e.g. `12461`), printed small at the top of the voucher. UBL does **not** need it to look up a bill. It is returned by the inquiry for reference, and UBL may echo it in the payment request (Section 7).

3 Voucher Resolution: Latest Voucher Rule & TAFS Consent

Every inquiry and payment made with a CC or a 1Bill consumer number resolves to **the most recent voucher generated by the school for that student**. Concretely, it is the newest voucher, excluding:

- vouchers the school has cancelled because a newer voucher replaced them; and
- vouchers the school has generated but not yet released to the parent.

Two active vouchers. If a student has more than one active (unpaid) voucher, the **most recent one is shown and collected first**. The older one is presented only once the most recent one has been paid. In normal use this is rare: when the school issues a new voucher, it carries the older unpaid amounts as arrears and cancels the older voucher.

If the student has no active voucher, the API reports the state of the most recent one (already paid, expired or written off) and collects nothing.

Statement of consent. The American Foundation School confirms that it has requested this method and consents to it: every search by CC number (or by 1Bill consumer number), on every UBL channel, returns the most recent voucher generated by the school for that student as defined above. TAFS is satisfied that this is the correct bill to present and collect. TAFS accepts that UBL bears no responsibility for a payment being applied to that voucher rather than to any earlier voucher of the same student. Every voucher generated by the school carries the full amount the student owes, including any arrears.

4 Amount Rules

Situation	Amount payable
On or before the due date	The voucher amount (`amount_within_due_date`).
After the due date (until validity date)	Voucher amount + late payment surcharge, normally **PKR 1,000** (`amount_after_due_date`).
After the validity date	Not payable. The school must issue a new voucher (response code `04`).
Channel = `ONEBILL`	The amounts above **+ the 1Bill charges**, borne by the payer. This document assumes **PKR 30** until UBL confirms the actual charges (see the open questions at the start).
OTC, app, internet banking	The amounts above, with no addition.

- The inquiry returns `amount_payable`, the exact amount to collect *today* on the channel stated in the request.

- The payment must carry exactly that amount. Partial payments and overpayments are rejected (code `06`).
- TAFS wants the **parent to bear the 1Bill charges**: they are added on top of the fee amount shown to the payer. On a 1Bill payment TAFS records only the fee amount against the student; the 1Bill charge is noted against the payment and never credited to the student's account.
- TAFS can change the 1Bill charge by configuration as soon as UBL confirms it; no code change is needed.
- Dates are evaluated in Pakistan Standard Time (UTC+05:00).

5 Connectivity & Authentication

5.1 Base URL

```
https://tafs-backend-production.up.railway.app/api/v1/ubl
```

TAFS will also provide a separate test (UAT) base URL with test students before go-live.

5.2 Authentication

Every request must carry two HTTP headers. Credentials are issued by TAFS over a secure channel and are never sent in the URL.

Header	Type	Description
`X-UBL-User`	string	Service user ID issued to UBL
`X-UBL-Password`	string	Password for that service user
`Content-Type`	string	`application/json` (payment requests)

A request with missing or wrong credentials receives response code `10`. All traffic is HTTPS (TLS 1.2 or higher). TAFS can additionally restrict access to UBL's outbound IP addresses on request.

5.3 HTTP status

A well-formed, authenticated request always receives **HTTP 200**. The business outcome is in `response_code`. A malformed request receives **HTTP 400** with a description of the invalid field, for example a missing `channel` or a non-numeric `consumer_number`.

6 API 1 – Bill Inquiry (GET)

```
GET /api/v1/ubl/bill-inquiry?consumer_number={CC or 1Bill no}&channel={channel}
```

6.1 Query parameters

Parameter	Type	Required	Description
`consumer_number`	digits	Yes	CC number, up to 5 digits (OTC, app, internet banking), or 15-digit 1Bill consumer number.
`channel`	enum	**Yes**	`ONEBILL`, `OTC`, `APP` or `INTERNET_BANKING`. Decides whether the 1Bill charge is added.

6.2 Successful response fields

Field	Type	Description
response_code	string	"00"
response_desc	string	"Success"
consumer_number	string	As sent
onebill_consumer_number	string	The student's 15-digit 1Bill number
cc	string	Student CC number, 5 digits zero-padded
student_name	string	Student's full name
gr_number	string	School registration number
campus	string	Campus name
class	string	Class
voucher_number	string	Voucher being collected (plain serial)
billing_month	string	Month(s) billed, e.g. "OCT 26"
voucher_status	string	UNPAID, OVERDUE or PARTIALLY_PAID
issue_date	string	YYYYMMDD
due_date	string	YYYYMMDD
validity_date	string	YYYYMMDD; not payable after this date
currency	string	"PKR"
amount_within_due_date	number	Payable on or before the due date (incl. channel fee)
amount_after_due_date	number	Payable after the due date (incl. late fee and channel fee)
late_fee	number	Late payment surcharge applicable after the due date
channel	string	As sent
channel_fee	number	1Bill charge for ONEBILL (assumed 30), else 0
amount_payable	number	**Exact amount to collect today**

6.3 Sample – 1Bill inquiry before the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=100000000008088&channel=ONEBILL
X-UBL-User: <issued by TAFS>
X-UBL-Password: <issued by TAFS>
```


```
{
  "response_code": "00",
  "response_desc": "Success",
  "consumer_number": "100000000008088",
  "onebill_consumer_number": "100000000008088",
  "cc": "08088",
  "student_name": "HAFSA DAHRI",
  "gr_number": "7001",
  "campus": "Gulistan-e-Jauhar Campus",
  "class": "AS",
  "voucher_number": "12461",
  "billing_month": "OCT 26",
  "voucher_status": "UNPAID",
  "issue_date": "20261001",
  "due_date": "20261010",
  "validity_date": "20261020",
```

```
  "currency": "PKR",
  "amount_within_due_date": 90030,
  "amount_after_due_date": 91030,
  "late_fee": 1000,
  "channel": "ONEBILL",
  "channel_fee": 30,
  "amount_payable": 90030
}
```

6.4 Sample – OTC inquiry after the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=08088&channel=OTC
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "cc": "08088",
  "voucher_number": "12461",
  "voucher_status": "OVERDUE",
  "amount_within_due_date": 90000,
  "amount_after_due_date": 91000,
  "late_fee": 1000,
  "channel": "OTC",
  "channel_fee": 0,
  "amount_payable": 91000,
  ...
}
```

6.5 Sample – failure

A failed inquiry returns only `response_code`, `response_desc` and, where a voucher was found, `voucher_number`:

```
{ "response_code": "03", "response_desc": "Voucher is already paid", "voucher_number
    ": "12461" }
```

7 API 2 – Bill Payment (POST)

```
POST /api/v1/ubl/bill-payment
Content-Type: application/json
```

7.1 Request body

Field	Type	Required	Description
`consumer_number`	digits	Yes	Same value used in the inquiry
`channel`	enum	**Yes**	`ONEBILL`, `OTC`, `APP`, `INTERNET_BANKING`
`payment_method`	enum	**Yes**	How the payer paid. See Section 7.2

Field	Type	Required	Description
`amount`	number	Yes	Amount collected from the payer, PKR. Must equal `amount_payable` (incl. the 1Bill charge on 1Bill)
`transaction_id`	string	Yes	UBL's unique transaction reference (max 100). Used to prevent duplicate posting
`transaction_datetime`	ISO 8601	Yes	When the payment was taken, with offset, e.g. `2026-10-05T11:00:00+05:00`
`voucher_number`	string	Advised	`voucher_number` from the inquiry; guards against a newer voucher issued in between
`branch_code`	string	OTC	UBL branch code where cash/cheque was received
`payer_bank_code`	string	1Bill	Code of the bank or wallet the payer paid from
`payer_bank_name`	string	1Bill	Name of that bank or wallet
`instrument_number`	string	Cheque / PO	Cheque or pay-order number. Required when `payment_method` is `CHEQUE` or `PAY_ORDER`

7.2 Mandatory payment method tracking

TAFS requires every payment to state **both** the channel it arrived through and the method the payer actually used, so that each receipt can be reconciled. A payment without them is rejected (HTTP 400).

`payment_method`	Meaning
`CASH`	Cash at a branch counter
`CHEQUE`	Cheque (send `instrument_number`)
`PAY_ORDER`	Pay order / demand draft (send `instrument_number`)
`ACCOUNT_TRANSFER`	Debit from the payer's bank account (app, internet banking, 1Bill)
`DEBIT_CARD`	Debit card
`CREDIT_CARD`	Credit card
`WALLET`	Mobile wallet (e.g. via 1Bill)
`OTHER`	Any other method; please tell TAFS which so it can be added

7.3 Successful response

Field	Type	Description
`response_code`	string	`"00"`
`response_desc`	string	`"Success"`
`transaction_id`	string	As sent
`tafs_receipt_id`	string	TAFS receipt number for this payment
`voucher_number`	string	Voucher the payment was applied to
`amount_booked`	number	Amount credited to the student (excludes the 1Bill fee)

Field	Type	Description
`channel_fee`	number	1Bill charge borne by the payer and retained by 1Bill (`0` on other channels)

7.4 Sample – 1Bill payment

```
POST /api/v1/ubl/bill-payment
{
  "consumer_number": "100000000008088",
  "channel": "ONEBILL",
  "payment_method": "ACCOUNT_TRANSFER",
  "amount": 90030,
  "transaction_id": "UBL20261005110000123",
  "transaction_datetime": "2026-10-05T11:00:00+05:00",
  "voucher_number": "12461",
  "payer_bank_code": "HBL",
  "payer_bank_name": "Habib Bank Limited"
}
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "transaction_id": "UBL20261005110000123",
  "tafs_receipt_id": "48215",
  "voucher_number": "12461",
  "amount_booked": 90000,
  "channel_fee": 30
}
```

7.5 Sample – OTC cheque payment after the due date

```
{
  "consumer_number": "08088",
  "channel": "OTC",
  "payment_method": "CHEQUE",
  "instrument_number": "00412387",
  "branch_code": "0457",
  "amount": 91000,
  "transaction_id": "UBLOTC0457202610121015",
  "transaction_datetime": "2026-10-12T10:15:00+05:00",
  "voucher_number": "12461"
}
```

7.6 Sample – failures

```
{ "response_code": "06", "response_desc": "Amount does not match the payable amount
    ", "amount_payable": 90030 }
{ "response_code": "08", "response_desc": "Duplicate transaction; already posted",
  "transaction_id": "UBL20261005110000123", "tafs_receipt_id": "48215" }
```

7.7 Duplicates, retries and timeouts

- Posting is idempotent on `transaction_id`. If UBL does not receive a response (timeout), it should resend the **same** request with the **same** `transaction_id`. A payment that was already posted returns code `08` with the original `tafs_receipt_id` and is never booked twice.
- Code `08` therefore means the payment **is** recorded at TAFS and should be treated as a success.
- Code `99` means nothing was booked; UBL may retry.

8 Response Codes

Code	Meaning	Used by
`00`	Success	Inquiry, Payment
`01`	Consumer number not found	Inquiry, Payment
`02`	No voucher has been issued for this student	Inquiry, Payment
`03`	Voucher is already paid	Inquiry, Payment
`04`	Voucher validity has expired; school must issue a new voucher	Inquiry, Payment
`05`	Voucher is not payable (written off)	Inquiry, Payment
`06`	Amount does not match the payable amount	Payment
`07`	A newer voucher has been issued; inquire again	Payment
`08`	Duplicate transaction; already posted (treat as success)	Payment
`09`	Invalid request (e.g. cheque without `instrument_number`)	Payment
`10`	Invalid credentials	Inquiry, Payment
`99`	General error; nothing booked, retry later	Inquiry, Payment

9 What Is Printed on the Voucher

Each voucher issued for collection into the UBL account carries, at the bottom:

- **UBL branches:** pay at any UBL branch by quoting the student's CC number.
- **UBL digital app:** go to Bill Payment, select "TAFS" as the beneficiary and enter the student's CC number.
- **Other banks / digital channels (1Bill):** pay via "1Bill Invoices" using the 1Bill consumer number printed on the voucher; 1Bill service charges are added to the bill and paid by the payer.

For The American Foundation School	For United Bank Limited
Name / Designation / Date	Name / Designation / Date